

TORCS Python Car on Windows – Troubleshooting Guide

This guide helps you get a Python-controlled car working in TORCS on Windows using the SCR server and a Python client (snakeoil-style). It prioritises a known-good baseline and Windows-specific fixes to shorten time-to-first-lap.

See our video for windows walkthrough here:

<https://www.youtube.com/watch?v=rRbpOUH0IFQ>

1. Baseline: Known- Good Setup Checklist

- Windows 10 or 11 (64-bit)
- Python 3.8–3.10 (avoid 3.11+ initially)

2. TORCS Will Not Start – Fixes

- Run TORCS once as Administrator; set Windows 7 compatibility
- Reset GPU overrides (NVIDIA/AMD control panels)

3. Python Client Cannot Connect

- In TORCS: Race → Quick Race → Configure Race → Driver = scr_server; then Start Race
- Match HOST=localhost, PORT=3001 in Python client
- Allow torcs.exe through Windows Defender Firewall

4. Immediate Crash When Race Starts

- Validate command ranges: steer $\in [-1,1]$, accel/brake $\in [0,1]$, gear integer
- Always send a minimal safe command each tick

5. Car Does Not Move

- Set gear=1 when speed<1
- accel > 0 and brake = 0
- Avoid clutch/auto gears at first

6. Steering Oscillation

- Dampen steering and rate-limit

7. Performance Too Slow

- Disable heavy visuals/logging

8. Windows-Specific Issues

- set PYTHONUTF8=1
- where python (resolve multiple versions)
- Use another port if 3001 is busy

Minimal safe command

```
command = {  
    'accel': 0.2,  
    'brake': 0.0,  
    'steer': 0.0,  
    'gear': 1  
}
```

Port/host in Python client

```
HOST = 'localhost'  
PORT = 3001
```

Simple straight-line test

```
steer = 0.0  
accel = 0.3  
brake = 0.0  
gear = 1
```

Troubleshooting Flowchart

Follow this high-level decision path to quickly isolate whether the fault is in the environment, network/socket, or driver logic.



